Osama Yousuf

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Professional Summary

- PhD Candidate in Machine Learning (ML) with 5+ years of experience in deep learning models, architectures, and frameworks (excelling in PyTorch), and 10+ publications & presentations on ML with new computing paradigms.
- Experienced in leading research projects end-to-end and presenting progress to both technical teams and non-technical stakeholders through detailed technical reports, presentations, and interactive demos.
- **Research interests:** Deep learning, neuromorphic computing, low-rank training, neural network quantization, hardware accelerators for Artificial Intelligence (AI), hardware-software co-design, Large Language Models (LLMs).

Work Experience

Graduate Research Assistant — *Washington, DC*
George Washington University — *Jan 2021 – May 2025*

Advised by Gina C. Adam at the ADAM Lab, focusing on robust neural network algorithms for noisy hardware accelerators.

Research Topic 1: Accurate Deep Neural Network Inference in Noisy Hardware

- Developed a fault-tolerant inference scheme for neural network accelerators based on noisy, imperfect hardware.
- Trained multi-layer perceptron networks (MLP) and convolutional neural networks (CNNs) and showed that the scheme improves performance by $\geq 150\%$ compared to prior methods under similar fault conditions (patent pending).

Research Topic 2: Resource-efficient Training of Deep Neural Networks

- Developed low-rank training algorithms for neural networks that enable in-situ training in noisy hardware from scratch. Unlike low-rank adaptation (LoRA) in LLMs, this method directly approximates backpropagated gradients in low-rank subspaces. Hardware MLP networks achieve near-software performance ($< 1\%$) with substantial memory savings.
- Assisted on the testing of *QSArray*, a novel hardware architecture based on systolic arrays for accelerating deep neural network training. Developed supporting code that enables high-level PyTorch networks (transformers, multi-layer perceptrons, etc.) to delegate gradient calculations to acceleration cores in C++.

Research Topic 3: Libraries & Toolkits for Deep Learning in New Computing Architectures

- Developed *NeuroTorch*, a deep learning library tailored for studying in-memory accelerators for deep neural networks. It supports diverse quantization schemes, fault-tolerance algorithms, and analog device models. NeuroTorch layers leverage fast pyBind11 bindings to offload backpropagation and inference calculations to custom C++ acceleration cores.
- Contributed to *Daffodil*, a mixed-signal hardware prototyping system for benchmarking accelerators for deep neural networks based on resistive memories. Implemented core abstraction modules for a high-level Python library that enables execution of neural network layers (training and inference) natively on the platform.

Machine Learning Research Intern — *San Jose, CA*
Western Digital Corporation — *Oct 2024 – Dec 2024*

Advised by Mike Grobis, Martin Lueker-Boden, and Michael Tran from *Western Digital Research*.

Research Topic: Inference Accelerators Large Language Models (LLMs)

- Developed a framework for benchmarking large language models (LLMs) under a variety of noise sources in analog AI accelerators. It extends PyTorch and integrates with *transformers* (by HuggingFace) and *lm-eval* for efficient language model evaluation. All operations are CUDA-compatible, with a minimal overhead of only $1.1\times$ in simulation time.
- Studied noise limitations for various state-of-the-art LLMs (Llama3, Phi, BitNet) and derived specifications for an analog LLM accelerator. For example, the accelerator must have device-to-device variability $\sigma \leq 15\%$ for attaining near-software performance ($< 3\%$ of baseline).

Machine Learning Research Intern — *College Station, TX*
Texas A&M University — *May 2019 – Aug 2019*

Hosted by Guni Sharon at the π^* AI & Optimization Lab.

Research Topic: Autonomous Vehicles using Reinforcement Learning (RL)

- Developed a Reinforcement Learning (RL) framework for the CARLA environment that enables sim-to-real policy transfer for autonomous vehicles. This was adopted by lab members to study RL algorithms for autonomous driving.
- Applied reward shaping, action constraints, and curriculum learning to enhance policy performance and ensure applicability to real-world driving scenarios. Trained a Deep Q-learning agent to control a vehicle using video inputs.
- Successfully transferred policies to the *DeepRacer* robot using a human-in-the-loop system that fine-tunes policies based on human expert demonstrations and feedback.

Education

George Washington University *Washington, DC*
PhD in Computer Engineering *Jan 2021 – May 2025*

- Focus: Machine Learning and Intelligent Systems | CGPA: 4.00/4.00, Transcript.
- **Dissertation:** Modeling, Prototyping, and Algorithm Design for *Memristive Neural Networks.*

Habib University *Karachi, Pakistan*
Bachelors in Computer Science & Mathematics *Aug 2016 – July 2020*

- Awards: Summa Cum Laude (Class Rank: 1) | CGPA: 3.92/4.00, Transcript.
- **Final Year Project:** Personal Outfit Recommendation System for eastern clothing (transfer learning).

Skills

Languages	Python, C++, Verilog, SQL, JavaScript
Frameworks	PyTorch, PyTorch Lightning, Scikit-Learn, Huggingface, Streamlit, Gradio, wandb, Gym, Numpy, Pandas
Visualization	Matplotlib, Altair, Seaborn, Wolfram Mathematica, MATLAB
Web Dev.	Django, Flask, HTML/CSS, REST APIs, Angular, WebGL
Prototyping	Vivado, PYNQ, cocotb, hls4ml, Uboot, Synopsys Design Vision, Waveforms, PSpice, FPGA Programming
Misc.	Jupyter Notebook, pyBind11, ctypes, Sphinx, Git, LaTeX

Patents & Publications

Layer Ensemble Averaging for Fault-Tolerant Hardware Neural Networks *2025*
O. Yousuf, G. C. Adam.
Provisional Patent, United States, 130761-00508, January, 2025.

Robust Hardware-Aware Neural Networks for FeFET-based Accelerators *2024*
O. Yousuf, A. L. Glasmann, A. L. Mazzoni, S. Najmaei, G. C. Adam.
Under review at *IEEE Transactions on Nanotechnology.*

Layer Ensemble Averaging for Fault Tolerance in Memristive Neural Networks *2024*
O. Yousuf, B. Hoskins, K. Ramu, M. Fream, W. A. Borders, A. Madhavan, M. W. Daniels, et al.
Nature Communications: Neuromorphic Computing, 10.1038/s41467-025-56319-6.

Neural Network Modeling Bias for Hafnia-based FeFETs *2023*
O. Yousuf, I. Hossen, A. L. Glasmann, S. Najmaei, G. C. Adam.
International Symposium on Nanoscale Architectures (NANOARCH), 10.1145/3611315, Slides.

Device Modeling Bias in ReRAM-based Neural Network Simulations *2023*
O. Yousuf, I. Hossen, M. W. Daniels, M. Lueker-Boden, A. Dienstfrey, G. C. Adam.
IEEE Journal on Emerging and Selected Topics in Circuits and Systems, 10.1109/JETCAS.2023.3238295.

Gradient Decomposition Methods for Training Networks with Non-Ideal Synapses *2021*
J. Zhao, S. Huang, **O. Yousuf**, Y. Gao, B. Hoskins, G. C. Adam.
Frontiers in Neuroscience: Neuromorphic Computing, 10.3389/fnins.2021.749811.

A System for Validating Resistive Neural Network Prototypes *2021*
B. Hoskins, W. Ma, M. Fream, **O. Yousuf**, Y. Gao, B. Hoskins, G. C. Adam.
International Conference on Neuromorphic Systems (ICONS), 10.1145/3477145.3477260.

Talks & Presentations

Accelerating Large Language Models with Magnetic RAM Crosspoint Arrays *Dec 2024*
Invited Talk at *Frontier Seminar Series, Western Digital Corporation, San Jose, CA.*

Layer Ensemble Averaging for Improving Memristive Neural Network Performance *Oct 2024*
Poster Presentation at *Innovation Bazaar, Western Digital, San Jose, CA.*

Daffodil: A Robust Prototyping System for Resistive Neural Networks *Nov 2023*
Poster Presentation at *NIST Prep Symposium, Gaithersburg, MD.*

Investigating Bias in the Modeling of Resistive RAM Devices *Dec 2022*
Poster Presentation at *International Conference on MEMRISYS, Cambridge, MA.*

Towards a Hardware-Aware Decomposition Method for ReRAM Neural Networks *Jul 2022*
Oral Presentation at *International Conference on Neuromorphic Systems (ICONS), Knoxville, TN.*

Streaming Gradient Tracking using Non-negative Matrix Factorization *Apr 2022*
Poster Presentation at *GW Research Showcase, George Washington University, Washington, DC.*